

Diaphorina Model

Auxiliary Functions

System Equations

SystemEvolve numerically integrates a system of nonlinear ordinary differential equations representing the population dynamics of a citrus agroecosystem, including *Diaphorina citri* (parasite), plant shoots, *Tamarixia radiata* (parasitoids), and external stimuli.

Periodic augmentative releases of parasitoids are implemented using WhenEvent, adding a fixed amount δ to the parasitoid population every T time units, while allowing the population to evolve continuously according to its intrinsic dynamics between releases.

The system is solved using NDSolve with an explicit Runge–Kutta time integration scheme, and the resulting population trajectories are visualized over the specified time horizon.

```
In[2]:= ClearAll[SystemEvolve];
SystemEvolve[δ_, T_, {{r1_, b12_, b13_, a1_, c1_}, {r2_, b21_, b24_, a2_, c2_}, {r3_, b31_, a3_, c3_}},
{x, eqs, x1, x2, x3, x4},
eqs=NDSolve[{
(x1'[t]==x1[t]*(r1+b12*x2[t]+b13*x3[t]-(a1+c1*(b12*x2[t]+b13*x3[t]))*x1[t])), (*Diaphorina citri*)
(x2'[t]==x2[t]*(r2+b21*x1[t]+b24*x4[t]-(a2+c2*(b21*x1[t]+b24*x4[t]))*x2[t])), (*Plant shoots*)
(x3'[t]==x3[t]*(r3+b31*x1[t]-(a3+c3*b31*x1[t])*x3[t])), (*Tamarixia radiata*)
(x4'[t]==x4[t]*(r4+b42*x2[t]-(a4+c4*b42*x2[t])*x4[t])), (*Stimuli*)
x1[0]==Di0,
x2[0]==B0,
x3[0]==x30,
x4[0]==x40,
WhenEvent[Mod[t,T]==0 && t>0, x3[t]→x3[t]+δ]
},
{x1,x2,x3,x4},{t,0,tt},Method->{"TimeIntegration"->"ExplicitRungeKutta"}];
Plot[Evaluate[{x1[t],x2[t],x3[t],x4[t]}/.eqs,{t,0,tt},
PlotLegends->{"Diaphorina citri (Parasite)","Plant shoots","Tamarixia radiata (Parasitoid)","Stimuli"},
PlotRange->All, PlotLabel->"Generalized model", AxesLabel->{"Time", "Population"}]
]
```

Initial Values and System Parameters

- Population:
 - x_1 : *Diaphorina citri* | x_2 : Plant shoots | x_3 : *Tamarixia radiata* | x_4 : Stimuli
- r_i : Population i intrinsic growth rate
- a_i : Population i saturation rate
- b_{ij} : Interaction between species
 - $b_{ij} > 0 \rightarrow$ Population i increases due to j

■ $b_{ij} < 0 \rightarrow$ Population i decreases due to j

■ c_i : Population i occupation rate

```
In[4]:= as = 0.000555; cs = 0.005;
param1 = {-0.2, 0.04, -0.01, as, cs};
param2 = {-0.72, -0.01, 0.054, as, cs};
param3 = {-0.3, 0.01, as, cs};
param4 = {0.72, -0.072, as, cs};
init = {50, 10, 15, 30, 100};
ChartReport[init, param1, param2, param3, param4]
```

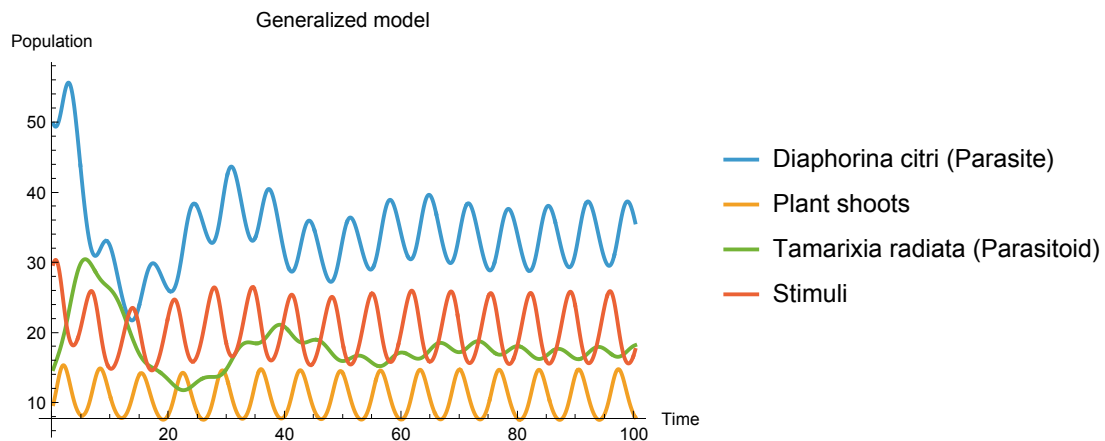
Out[10]=

init		x1		x2		x3		x4
x1[0]	50	r1	-0.2	r2	-0.72	r3	-0.3	r4
x2[0]	10	b12	0.04	b21	-0.01	b31	0.01	b4
x3[0]	15	b13	-0.01	b24	0.054	a3	0.000555	a4
x4[0]	30	a1	0.000555	a2	0.000555	c3	0.005	c4
t	100	c1	0.005	c2	0.005			

No Periodic Parasitoid Release

```
In[*]:= SystemEvolve[0, 1, {param1, param2, param3, param4}, init]
```

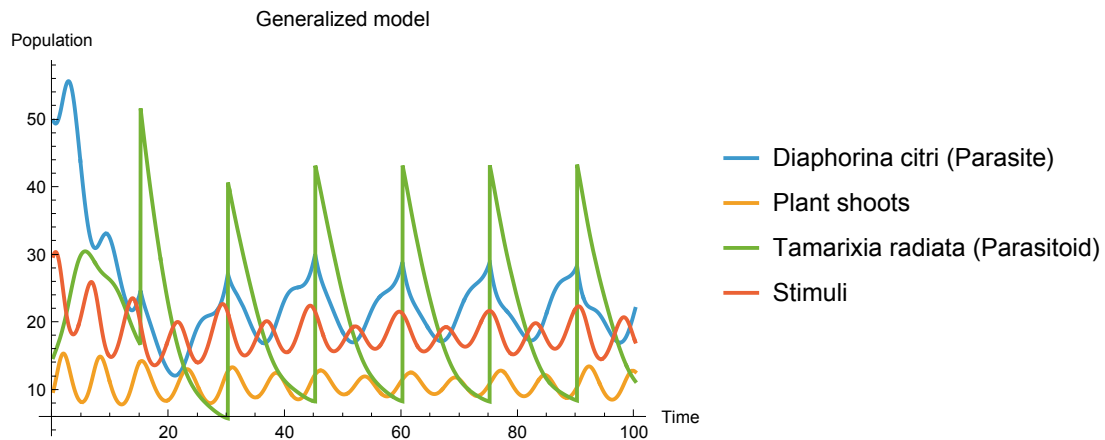
Out[*]=



Periodic Parasitoid Release

```
In[ ]:= SystemEvolve[35, 15, {param1, param2, param3, param4}, init]
```

```
Out[ ]:=
```



Interactive Parameter Exploration

```
In[ ]:= Manipulate[
  SystemEvolve[ $\delta$ , T, {param1, param2, param3, param4}, {50, 10, 15, 30, t}],
  {{ $\delta$ , 35}, 0, 100, 1},
  {{T, 15}, 0, 100, 1},
  {{t, 100}, 1, 1000, 10}
]
```

```
Out[ ]:=
```

